

Arithmetic Convolution Factors of Fabius-Type Laws

Exact divisibility, fractal remainders,
and classification barriers

Prepared for the ProveIt research programme

Abstract

Let $A_0 = 1$ and let A_k be a strictly increasing chain of positive integers under divisibility. The random series $X_A = \sum_{k \geq 0} U_k/A_k$, with independent centered unit uniforms U_k , has a compactly supported C^∞ density. We give an exact classification of factorizations $\mu_A = D_c \mu_B * \nu$ by probability measures: they exist precisely when $c = 1/m$ for a positive integer m and $A_k \mid mB_k$ for every k . The unique remainder has an explicit independent-digit representation. This converts a positive-definiteness problem for entire Fourier quotients into coordinatewise arithmetic.

The classification yields a rigid cross-base theorem for geometric-uniform laws, an embedding of inclusion modulo finite sets into scaled convolution factorability, and a compact family of smooth convolution divisors of the single Rvachev law on which mutual factorability has no Borel real-valued complete invariant. With primitive-recursive ladder presentations, fixed-scale factorability is Π_1^0 -complete and existence of some scale is Σ_2^0 -complete. Explicit spline approximation bounds show that these barriers persist for uniformly computable smooth densities. For same-base geometric laws we also classify the remainder as singular continuous, a finite uniform spline, or a finite smoothing of a singular measure; we prove sharp integer regularity, sharp power Fourier decay, and a sharp Hölder result in the separated-digit case. Further questions isolate the boundaries of the classification rather than presuming that they are already solved.

Status and scope. All asserted mathematical results below have written proofs. The auxiliary program performs exact finite regression checks, not proof-assistant verification. No Lean verification or peer review is claimed. Classical inputs and identified repository overlaps are distinguished from the extensions developed here. Priority of the latter has not been established by an exhaustive literature or repository audit.

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1 Research question, provenance, and main conclusions

1.1 The question behind the Fourier product

The Rvachev up-function is a particularly structured compactly supported smooth probability density. Its Fourier transform is an infinite product of sinc functions. In the normalization used here,

$$\widehat{\text{up}}(t) = \prod_{k=0}^{\infty} \frac{\sin(\pi t/2^k)}{\pi t/2^k}. \quad (1)$$

Arias de Reyna gives this product and the integer-zero multiplicity formula $1 + v_2(n)$ explicitly [2]. These formulas, and the probabilistic construction underlying them, are established starting points, not new claims of this article. Fabius’s original construction provides the historical probabilistic context [1].

The product suggests a question that is more restrictive than formal power-series division. Given two such laws, when does removing a scaled copy of one from the other leave a *probability measure*? Precisely, for which $c > 0$ is there a probability ν such that

$$\mu = D_c \rho * \nu? \quad (2)$$

Here $D_c \rho$ is the law of cY when Y has law ρ , and $*$ is convolution, equivalently addition of independent random variables. A formally defined Fourier quotient need not be a characteristic function. Even an entire quotient is not automatically positive definite. The central result is that for a natural class of nested integer denominators, these analytic and positivity questions have a complete arithmetic answer.

1.2 Connection with the repository and limits of the audit

The repository was inspected at commit `fbba58593dc0622aa914972896150d4848f935b5`. Its Fabius research documentation discusses geometric-uniform laws, Fourier and Laplace products, moment reciprocity, and deconvolution [5]. Its Shape, Divisibility, and Stein Geometry package already includes identical-convolution rootlessness for the Fabius–Rvachev law [6]. We therefore do not present the dyadic zero formula, probabilistic construction, or dyadic rootlessness as repository-new results.

Our principal extension is to two arbitrary divisibility ladders, and then to the order-theoretic and effective complexity encoded by those ladders. The same-base remainder analysis supplies a complementary analytic classification. The general geometric self-factor theorem is included as a useful bridge and is proved independently; no priority claim is attached to that elementary product consequence.

The audit was scoped, not exhaustive. In particular, direct retrieval of the large geometric- q frontier TeX file exceeded the connector’s supported response size. Navigation records and smaller relevant files were inspected, but absence of a theorem from those records is not evidence that it occurs nowhere in the larger corpus. The article thus establishes mathematical statements and gives their proofs, without claiming to settle a named long-standing conjecture whose open status has not been verified. The classical descriptive-set-theoretic obstruction used later is also not new: the contribution there is its explicit realization by these smooth laws, not the nonsmoothness of finite-change equivalence itself [4].

1.3 What is resolved here

The first classification is the following exact criterion. Suppose $A_0 = B_0 = 1$, $A_k \mid A_{k+1}$ and $B_k \mid B_{k+1}$, with both chains strictly increasing. Then

$$\mu_A = D_c \mu_B * \nu \iff \begin{cases} c = 1/m \text{ for an integer } m \geq 1, \\ A_k \mid mB_k \text{ for every } k \geq 0. \end{cases} \quad (3)$$

This is Theorem 3.2. The implication from left to right uses zero multiplicities; the reverse implication constructs positive digit measures and therefore resolves, rather than assumes, positive definiteness.

Several consequences make the criterion more than an isolated product identity. An integer-base target can contain a geometric-uniform source only when the reciprocal source ratio is an integer multiple of the target base (Theorem 4.2). A single compact family of smooth densities realizes inclusion modulo finite sets (Theorem 5.1) and finite-change equivalence (Theorem 5.4). Exact effective classification has specified arithmetical-hierarchy complexity even though the densities themselves are uniformly approximable with every derivative (Theorem 6.1). Finally, the explicit remainders exhibit a sharp division between singular measures and finite levels of smoothing (Theorem 7.4).

The results concern genuine independent convolution factors. They do not classify signed deconvolution kernels, arbitrary nonlinear transports, approximate factorizations, or every compactly supported smooth law.

2 Probability laws, entire products, and finite approximations

2.1 Definitions and normalization

Throughout, $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and $\mathbb{N}_{>0} = \{1, 2, \dots\}$. Let U_L denote uniform probability on $[-L/2, L/2]$, for $L > 0$. Let $(U_k)_{k \geq 0}$ be independent random variables with law U_1 . Our Fourier convention and normalized sinc are

$$\hat{\sigma}(t) = \int_{\mathbb{R}} e^{2\pi i t x} d\sigma(x), \quad \text{sinc } z = \begin{cases} \sin(\pi z)/(\pi z), & z \neq 0, \\ 1, & z = 0. \end{cases}$$

All laws considered as targets are symmetric, so the opposite Fourier sign gives the same products.

Definition 2.1 (Divisibility ladder). A divisibility ladder is a sequence $A = (A_k)_{k \geq 0}$ of positive integers such that $A_0 = 1$, $A_k \mid A_{k+1}$, and $A_{k+1} > A_k$. Set

$$X_A = \sum_{k \geq 0} \frac{U_k}{A_k}, \quad \mu_A = \text{Law}(X_A), \quad R_A = \frac{1}{2} \sum_{k \geq 0} A_k^{-1}. \quad (4)$$

For a real ratio $0 < q < 1$, separately set

$$X_q = \sum_{k \geq 0} q^k U_k, \quad \mu_q = \text{Law}(X_q). \quad (5)$$

For an integer $a \geq 2$, abbreviate $\mu^{[a]} = \mu_{1/a}$; this is the ladder law for $A_k = a^k$. In particular, $\mu^{[2]}$ has density up in (1), and is not its cumulative distribution function.

The ladder ratios are integers at least two, so $A_k \geq 2^k$. Both random series converge absolutely for every choice of the bounded coordinates. In the ladder case $R_A \leq 1$; in the geometric case the support radius is $1/(2(1-q))$.

Lemma 2.2 (Transforms, support, and smoothness). *For a divisibility ladder,*

$$\Phi_A(z) := \hat{\mu}_A(z) = \prod_{k \geq 0} \text{sinc}(z/A_k) \quad (6)$$

converges locally uniformly on $\mathbb{C}$, defines an entire function, and has no zeros other than the zeros of its individual factors. The law μ_A has a C^∞ density f_A with support $[-R_A, R_A]$. The same assertions hold for μ_q , with factors $\text{sinc}(q^k z)$ and support $[-1/(2(1-q)), 1/(2(1-q))]$.

Proof. Uniform convergence of the random series implies convergence of its characteristic functions by bounded convergence. On a compact disk, $\text{sinc}(z/A_k) = 1 + O(A_k^{-2})$ uniformly, and $\sum A_k^{-2} < \infty$. The usual product criterion follows directly by taking the analytic logarithm of sufficiently late factors: their deviations from one have summable suprema. Thus the tail product is analytic and nonzero on each small disk avoiding zeros of the finitely many initial factors. This proves both local uniform convergence and the assertion about zeros.

For every positive integer M and real $|t| \geq 1$,

$$|\Phi_A(t)| \leq \prod_{k=0}^{M-1} \frac{A_k}{\pi|t|} = \frac{\prod_{k < M} A_k}{\pi^M |t|^M}, \quad (7)$$

since the other factors have absolute value at most one. Consequently $t^r \Phi_A(t)$ is integrable for every $r \geq 0$. Fourier inversion and termwise differentiation under the integral give a smooth density.

The series is confined to $[-R_A, R_A]$. For completeness its support is the entire interval. The possible sums of the first $N+1$ support intervals form $[-R_{A,N}, R_{A,N}]$, where $R_{A,N} \uparrow R_A$. The tail has zero in its support: its first finitely many variables can be made arbitrarily small with positive probability, and its remaining absolute bound tends to zero. Any interior point of $[-R_A, R_A]$ therefore lies in the support of a sufficiently long prefix plus its independent tail. Taking closures includes the endpoints. Replace A_k^{-1} by q^k for the geometric case; all arguments are unchanged. $\square$

Lemma 2.3 (Zero multiplicities). *The zero set of Φ_A is $\mathbb{Z} \setminus \{0\}$, and for $n \neq 0$,*

$$\text{ord}_{z=n} \Phi_A(z) = v_A(n) := \#\{k \geq 0 : A_k \mid n\}. \quad (8)$$

The zero at 1 is simple. For a geometric law, the zero set is $\{nq^{-k} : n \in \mathbb{Z} \setminus \{0\}, k \geq 0\}$, and the zero at 1 is again simple.

Proof. The k th ladder factor vanishes simply exactly at nonzero multiples of A_k . Only finitely many factors vanish at a fixed point. Their orders add, and the remaining product is nonzero by Lemma 2.2. The first factor accounts for every nonzero integer, and later factors produce no noninteger zeros. Only the first factor vanishes at 1. The same argument applies to the geometric factors; $0 < q^k < 1$ for $k > 0$ prevents any additional vanishing at 1. $\square$

2.2 Why compactness makes zero multiplicities legitimate

Lemma 2.4 (Compact factor and cancellation). *Suppose compactly supported probability measures μ, ρ satisfy $\mu = \rho * \nu$ for a probability measure ν . Then ν is compactly supported, the Fourier identity extends to entire functions, and*

$$\text{ord}_z \hat{\mu} \geq \text{ord}_z \hat{\rho}$$

at every zero of $\hat{\rho}$. If $\hat{\rho}$ is nonzero except at isolated real points, the factor ν is unique. If μ and ρ are symmetric, this unique factor is symmetric.

Proof. For support points $x \in \text{supp } \rho$ and $y \in \text{supp } \nu$, every neighborhood of $x + y$ contains the sum of two positive-measure neighborhoods; independence therefore gives $x + y \in \text{supp } \mu$. Fixing one x bounds every possible y , so ν has bounded closed support. The transforms are entire and agree on the real axis, hence everywhere. Orders of zeros add under multiplication.

For uniqueness, divide the characteristic functions where $\widehat{\rho}(t) \neq 0$. The resulting equality for two candidate factors extends to every real t by continuity, and Fourier uniqueness gives equality of measures. Reflection of a factor is another factor when both target and source are symmetric, proving symmetry by uniqueness. The same support argument, applied to minimum and maximum points of compact supports, shows that the extreme support points add under convolution. $\square$

Corollary 2.5 (Identical-convolution rootlessness). *None of the laws μ_A or μ_q is ρ^{*j} for a probability measure ρ and an integer $j \geq 2$.*

Proof. A putative root is compact by the support argument. Its entire transform raised to the j th power would have a zero at 1 of order divisible by j , contradicting Lemma 2.3. $\square$

The dyadic instance of this short obstruction is already covered by the repository's divisibility report [6]. We include the uniform extension only to delimit the meaning of factorization here: nonidentical factors can exist even though identical roots do not.

2.3 Explicit approximation in every derivative

Proposition 2.6 (Effective spline approximation). *Let $f_{A,N}$ be the density of $\sum_{k=0}^N U_k/A_k$. For $r \geq 0$ and $N \geq r + 2$, put*

$$L_r(A) = 2^{r+1} \prod_{j=0}^{r+1} A_j.$$

Then

$$\|f_A^{(r)} - f_{A,N}^{(r)}\|_\infty \leq \frac{L_r(A)}{2A_N}. \quad (9)$$

Every computable ladder therefore yields a computable C^∞ density, with a modulus supplied explicitly for each derivative.

Proof. Let g be the density of the first $r + 3$ uniforms, indexed 0 through $r + 2$. The distributional derivative of the density of U_{1/A_j} is $A_j(\delta_{-1/(2A_j)} - \delta_{1/(2A_j)})$, with total variation $2A_j$. Differentiate the factors $j = 0, \dots, r$, leaving two uniform factors undifferentiated. Their convolution is continuous and has sup norm at most A_{r+1} . It follows that $g^{(r+1)}$ is continuous with norm at most $L_r(A)$, and $g^{(r)}$ is globally $L_r(A)$ -Lipschitz. Convolution with any remaining prefix preserves this Lipschitz bound.

The tail $T_N = \sum_{k>N} U_k/A_k$ obeys

$$|T_N| \leq \frac{1}{2} \sum_{h \geq 1} \frac{1}{A_{N+h}} \leq \frac{1}{2A_N}.$$

Since $f_A^{(r)}(x) = \mathbb{E}f_{A,N}^{(r)}(x - T_N)$, the Lipschitz estimate gives (9).

To make computability explicit, write $\ell_j = 1/A_j$ and $S_N = \sum_{j=0}^N \ell_j$. The prefix has the finite spline formula

$$f_{A,N}(x) = \frac{1}{N! \prod_{j=0}^N \ell_j} \sum_{E \subseteq \{0, \dots, N\}} (-1)^{|E|} \left(x + \frac{S_N}{2} - \sum_{j \in E} \ell_j \right)_+^N. \quad (10)$$

This follows by repeated convolution of interval indicators, or by applying the corresponding finite differences to $x_+^N/N!$. It is a rational piecewise polynomial with computable knots. Choose N until the right side of (9) is smaller than the requested error; this search terminates because $A_N \geq 2^N$. Differentiating the rational spline gives uniform approximants for each prescribed derivative. No efficiency claim is made for the exponential-size subset formula. $\square$

3 A complete classification of ladder factors

3.1 The positive splitting identity

Lemma 3.1 (One uniform splits into a smaller uniform and digits). *For an integer $M \geq 1$ and $L > 0$, let $\ell = L/M$. Then*

$$\mathbf{U}_L = \mathbf{U}_\ell * \left(\frac{1}{M} \sum_{j=0}^{M-1} \delta_{\ell(j-(M-1)/2)} \right). \quad (11)$$

Proof. The translated intervals of length ℓ are adjacent and tile $[-L/2, L/2]$, up to endpoints of Lebesgue measure zero. On each interval the mixture density is $(1/M)(1/\ell) = 1/L$. This proves the probability identity directly, including $M = 1$ when the digit law is δ_0 . $\square$

3.2 The main theorem

Theorem 3.2 (Arithmetic convolution classification). *Let A and B be divisibility ladders and let $c > 0$. There is a probability measure ν satisfying*

$$\mu_A = D_c \mu_B * \nu$$

if and only if there is an integer $m \geq 1$ with

$$c = \frac{1}{m}, \quad A_k \mid mB_k \quad (k \geq 0). \quad (12)$$

When these conditions hold, the unique factor is

$$\nu_{A,B,m} = \text{Law} \left(\sum_{k \geq 0} \frac{J_k - (M_k - 1)/2}{mB_k} \right), \quad M_k = \frac{mB_k}{A_k}, \quad (13)$$

where the J_k are independent and uniform on $\{0, \dots, M_k - 1\}$. It is symmetric and has extreme support points $\pm(R_A - R_B/m)$.

Proof. Assume a factor exists. Its compactness and the entire Fourier identity follow from Lemma 2.4. The source transform $\Phi_B(cz)$ vanishes at $z = 1/c$. Every target zero is a nonzero integer, so $1/c = m$ is a positive integer.

Fix $k \geq 0$ and evaluate zero orders at $z = mB_k$. In the source,

$$\text{ord}_{z=mB_k} \Phi_B(z/m) = v_B(B_k) = k + 1,$$

because the divisors B_j of B_k are exactly those with $j \leq k$. The target must have order at least $k + 1$. For a divisibility chain, $\{j : A_j \mid mB_k\}$ is an initial segment. Having at least $k + 1$ elements is therefore equivalent to $A_k \mid mB_k$. This proves all conditions in (12).

Conversely suppose those conditions hold. The integers M_k in (13) are positive. Apply Lemma 3.1 to the target uniform of length $1/A_k$, with small length $1/(mB_k)$ and digit count

M_k . Construct all small uniforms and all digits independently. Each target summand is then the sum of one small uniform and one centered digit, and these target summands are independent across k .

The digit series converges absolutely, since its k th term has absolute value at most

$$\frac{M_k - 1}{2mB_k} = \frac{1}{2} \left(\frac{1}{A_k} - \frac{1}{mB_k} \right),$$

and the sum of these bounds is finite. Summing the independent small uniforms gives law $D_{1/m}\mu_B$; summing the digits gives (13). This proves the factorization by an actual probability measure, without an appeal to a merely formal quotient.

Uniqueness and symmetry follow from Lemma 2.4. Its support extrema must add to those of the source to give $\pm R_A$, so they are $\pm(R_A - R_B/m)$. This conclusion concerns extreme points; the intervening support need not be an interval. $\square$

Remark 3.3 (Why the indexing matters). The argument would fail for a general unordered list of integer frequencies: zero counts need not then form initial segments. The ladder hypothesis is the mechanism that converts a zero-order inequality into a coordinatewise divisibility condition. Positivity is supplied separately by interval tiling. Neither step may be omitted in an attempted extension to arbitrary products.

Corollary 3.4 (Equivalent zero-order criterion). *For $m \geq 1$, the factor in Theorem 3.2 exists if and only if*

$$v_A(mn) \geq v_B(n) \quad (n \geq 1). \quad (14)$$

It is enough to test the generator frequencies $n = B_k$.

Proof. Necessity follows from the entire factorization. If all generator inequalities hold, the proof of Theorem 3.2 gives the coordinatewise criterion. Conversely, if $B_j \mid n$ then $A_j \mid mB_j \mid mn$, proving (14). $\square$

3.3 The scale set is controlled by a single lcm obstruction

Definition 3.5 (Finite obstruction lcm). Put $d_k = A_k / \gcd(A_k, B_k)$. If the integers d_k have a common positive integer multiple, let $L(A, B)$ be their least common multiple. Otherwise write $L(A, B) = \infty$; the symbol ∞ here means absence of an ordinary finite integer common multiple, not a permitted scale parameter.

Corollary 3.6 (All scales and a largest one). *If $L(A, B) < \infty$, all possible positive scales are exactly*

$$\left\{ \frac{1}{L(A, B)r} : r \in \mathbb{N}_{>0} \right\}. \quad (15)$$

If $L(A, B) = \infty$, there is no possible positive scale. Equivalently, finite existence requires a positive integer whose prime exponents dominate

$$\lambda_p = \sup_{k \geq 0} \max\{0, v_p(A_k) - v_p(B_k)\} \quad (16)$$

for every prime p . Thus every λ_p must be finite and only finitely many of them may be positive; then $L(A, B) = \prod_p p^{\lambda_p}$.

Proof. The elementary equivalence $A_k \mid mB_k$ if and only if $A_k / \gcd(A_k, B_k) \mid m$ reduces Theorem 3.2 to the definition of the least common multiple. Prime factorization gives the last assertion. Both finiteness requirements are essential: bounded deficits at each prime do not help when infinitely many different primes are needed. $\square$

A failed fixed-scale factorization has a finite certificate: a single index k and prime p for which $v_p(A_k) > v_p(m) + v_p(B_k)$. The real number mB_k is then an explicit analytic obstruction, because the source has too high a zero multiplicity there. In contrast, confirming the criterion for arbitrarily many initial indices does not certify its truth for all indices. Section 6 makes this distinction exact.

Proposition 3.7 (Recovery and composition). *The law μ_A determines its ladder, by*

$$A_k = \min\{n \geq 1 : v_A(n) \geq k + 1\}. \quad (17)$$

If $A_k \mid mB_k$ and $B_k \mid nC_k$ for every k , then

$$\nu_{A,C,mn} = \nu_{A,B,m} * \mathbf{D}_{1/m} \nu_{B,C,n}. \quad (18)$$

Proof. The integers with at least $k + 1$ zero factors are exactly the positive multiples of A_k , proving (17). For composition, substitute $\mu_B = \mathbf{D}_{1/n} \mu_C * \nu_{B,C,n}$ into the factorization of μ_A . The resulting remainder must be the unique factor for scale $1/(mn)$, by Theorem 3.2. $\square$

3.4 Moment information carried by the factor

Let B_{2j} denote the usual even Bernoulli numbers, defined by $z/(e^z - 1) = \sum B_n z^n/n!$ near zero. For a compact law, its cumulants are the derivatives at zero of the logarithm of its moment generating function.

Proposition 3.8 (Exact cumulants). *For every $j \geq 1$,*

$$\kappa_{2j}(\mu_A) = \frac{B_{2j}}{2j} \sum_{k \geq 0} A_k^{-2j}, \quad (19)$$

$$\kappa_{2j}(\nu_{A,B,m}) = \frac{B_{2j}}{2j} \left(\sum_{k \geq 0} A_k^{-2j} - m^{-2j} \sum_{k \geq 0} B_k^{-2j} \right). \quad (20)$$

All odd cumulants vanish. In particular, $\text{Var}(X_A) = \frac{1}{12} \sum A_k^{-2}$.

Proof. For a centered unit uniform the moment generating function is $\sinh(z/2)/(z/2)$. Differentiating its logarithm gives $\frac{1}{2} \coth(z/2) - 1/z$, whose power series integrates to

$$\log \frac{\sinh(z/2)}{z/2} = \sum_{j \geq 1} \frac{B_{2j}}{2j(2j)!} z^{2j}.$$

Substitute z/A_k and sum near zero. Normal convergence justifies the coefficient calculation. Cumulants add under independent convolution and scale homogeneously, yielding (20). $\square$

3.5 A quantitative obstruction, without a bound on the remainder

For probability laws with finite first moments, write $W_1(\sigma, \tau)$ for the infimum of $\mathbb{E}|X - Y|$ over all couplings with these marginal laws. A simple characteristic-function inequality already gives a useful support-independent obstruction.

Proposition 3.9 (A Fourier lower bound for approximate factors). *For probability laws μ, ρ and every real $t \neq 0$,*

$$W_1(\mu, \rho * \nu) \geq \frac{(|\widehat{\mu}(t)| - |\widehat{\rho}(t)|)_+}{2\pi|t|} \quad (21)$$

for every probability remainder ν for which the left side is finite. Suppose in particular that $A_k \nmid mB_k$, put $t_0 = mB_k$ and $r = v_A(t_0) < k + 1$, and define

$$a_r = \frac{|\Phi_A^{(r)}(t_0)|}{r!} > 0, \quad D_r = \frac{(2\pi R_A)^{r+1} + (2\pi R_B/m)^{r+1}}{(r+1)!}. \quad (22)$$

For every $0 < h \leq a_r/(2D_r)$ and every probability ν ,

$$W_1(\mu_A, D_{1/m}\mu_B * \nu) \geq \frac{a_r h^r}{4\pi(t_0 + h)} > 0, \quad (23)$$

interpreting the bound trivially if the distance is infinite.

Proof. For a coupling, the inequality $|e^{2\pi i t x} - e^{2\pi i t y}| \leq 2\pi |t| |x - y|$ gives $|\widehat{\mu}(t) - \widehat{\rho * \nu}(t)| \leq 2\pi |t| W_1(\mu, \rho * \nu)$. Also $|\widehat{\rho * \nu}(t)| \leq |\widehat{\rho}(t)|$ because a probability characteristic function has modulus at most one. The triangle inequality proves (21).

At t_0 the target has order r , while the source has order at least $r + 1$. A compact law of radius R has Fourier derivatives bounded on the real axis by $(2\pi R)^j$ in order j . Taylor's formula with an integral remainder therefore gives

$$|\Phi_A(t_0 + h)| - |\Phi_B((t_0 + h)/m)| \geq a_r h^r - D_r h^{r+1} \geq \frac{1}{2} a_r h^r.$$

Apply (21) at $t = t_0 + h$ to obtain the claimed bound. No derivative, support, or moment estimate for ν was needed. $\square$

The leading coefficient in this bound can itself be written explicitly:

$$a_r = t_0^{-r} \prod_{j \geq r} |\operatorname{sinc}(t_0/A_j)|. \quad (24)$$

Indeed, each of the r vanishing factors has first derivative of absolute value $1/t_0$ at t_0 , and the remaining product is nonzero. Thus an arithmetic failure yields not only an exact nonexistence witness but a strict lower bound against *every* independent probability remainder. The bound may be very small for a high-index obstruction; no uniform separation over all failed instances is implied.

4 Geometric laws and cross-base rigidity

4.1 All self-factoring contractions, for every real ratio

Theorem 4.1 (Geometric self-factor spectrum). *For $0 < q < 1$,*

$$\{c > 0 : \mu_q = D_c \mu_q * \nu \text{ for some probability } \nu\} = \{q^j/m : j \in \mathbb{N}_0, m \in \mathbb{N}_{>0}\}. \quad (25)$$

Proof. If the factor exists, its compactness forces the zero $1/c$ of the scaled source to be a zero of the target. Lemma 2.3 gives $1/c = m q^{-j}$ with $m \geq 1$ and $j \geq 0$.

For the converse, first split the series at index j :

$$\mu_q = (U_1 * U_q * \cdots * U_{q^{j-1}}) * D_{q^j} \mu_q,$$

where the first convolution is δ_0 when $j = 0$. Split every uniform in the tail using the same integer digit count m , by Lemma 3.1. The smaller uniforms sum to law $D_{q^j/m} \mu_q$, and the initial block and all the independent digit terms form the probability remainder. Absolute convergence follows from the geometric bound. $\square$

This spectrum is countable and has no accumulation point other than zero: for a fixed $\varepsilon > 0$, $q^j/m \geq \varepsilon$ bounds both j and m . In particular, these laws do not admit such a self-factor decomposition at every contraction $0 < c < 1$. For integer reciprocal $q = 1/a$, the spectrum simplifies to all reciprocal positive integers, consistent with Theorem 3.2 for $A = B$.

4.2 An integer-base target forces an integer-base source

Theorem 4.2 (Cross-base classification). *Let $a \geq 2$ be an integer, $0 < q < 1$, and $c > 0$. Then*

$$\mu^{[a]} = D_c \mu_q * \nu \quad (26)$$

for a probability ν if and only if

$$q = \frac{1}{b} \text{ for an integer } b \geq 2, \quad a \mid b, \quad c = \frac{1}{m} \text{ for an integer } m \geq 1. \quad (27)$$

When this holds, the digit count in (13) is $M_k = m(b/a)^k$.

Proof. The zero at $1/c$ first gives $c = 1/m$. Every zero mq^{-k} of the source must be an integer zero of the target. Write $b = 1/q$. Since mb is an integer, b is rational. In lowest terms let $b = u/v$ with positive coprime integers u, v . The integrality of mu^k/v^k for every k forces $v^k \mid m$ for every k , which is impossible unless $v = 1$. Thus b is an integer at least two.

Theorem 3.2 now says $a^k \mid mb^k$ for every k . For each prime this is $kv_p(a) \leq v_p(m) + kv_p(b)$. Letting k increase forces $v_p(a) \leq v_p(b)$, hence $a \mid b$. Conversely that divisibility makes $a^k \mid mb^k$ automatic, and the explicit construction applies. $\square$

Corollary 4.3 (The Rvachev target). *A geometric-uniform law can be a positively scaled convolution factor of $\mu^{[2]}$ exactly when its ratio is $1/b$ with b even. The permitted scales are precisely $1/m$, $m \geq 1$. In particular, bases 6 and 10 are allowed, not only powers of 2; base 3 and all irrational ratios are excluded.*

Example 4.4 (Support and variance do not decide factorability). Consider the target $\mu^{[3]}$, the source $\mu^{[2]}$, and scale $c = 1/2$. The target has support radius $3/4$ and variance $3/32$; the scaled source has radius $1/2$ and variance $1/36$. These necessary size inequalities are satisfied. Nevertheless, at $t = 4$ the source transform $\Phi_{(2^k)}(t/2)$ has a zero of order 2, whereas $\Phi_{(3^k)}(t)$ has order 1. No probability remainder exists. This obstruction survives arbitrarily strong shrinking by changing the frequency at which the contradiction is read; the failure $3 \nmid 2$ rules out every positive scale.

Example 4.5 (A non-dyadic factor of the dyadic law). For target base 2, source base 6, and $m = 1$, the complement is

$$\text{Law} \left(\sum_{k \geq 0} \frac{J_k - (3^k - 1)/2}{6^k} \right), \quad J_k \text{ uniform on } \{0, \dots, 3^k - 1\}.$$

The $k = 0$ digit is deterministic. The support radius of this remainder is $1 - 3/5 = 2/5$. The theorem proves its existence and uniqueness, but does not by itself determine its absolute continuity or exact smoothness.

5 A compact smooth realization of inclusion modulo finite sets

5.1 Encoding one bit by one prime

Let $p_1, p_2, \dots$ list the odd primes in increasing order. For $S \subseteq \mathbb{N}_{>0}$ define

$$A_{S,0} = 1, \quad A_{S,k} = 2^k \prod_{\substack{1 \leq j \leq k \\ j \in S}} p_j \quad (k \geq 1), \quad \mu_S = \mu_{A_S}. \quad (28)$$

Each ratio is either 2 or $2p_{k+1}$, so these are divisibility ladders. Every μ_S consequently has a smooth density f_S supported in $[-1, 1]$. In fact, the whole family consists of unscaled convolution divisors of the single classical Rvachev law $\mu^{[2]}$.

Write $S \subseteq^* T$ when $S \setminus T$ is finite, and $S E_0 T$ when $S \Delta T$ is finite. The symbol E_0 denotes finite-change equivalence on binary sequences; its classical role in Borel classification is discussed in [4]. We will prove the obstruction needed here directly.

Theorem 5.1 (Exact arithmetic encoding). *For $S, T \subseteq \mathbb{N}_{>0}$ and $m \geq 1$,*

$$\mu_S = D_{1/m} \mu_T * \nu \iff S \setminus T \text{ is finite and } \prod_{j \in S \setminus T} p_j \mid m. \quad (29)$$

In particular, an unscaled factorization exists exactly when $S \subseteq T$; a factorization at some positive scale exists exactly when $S \subseteq^ T$. Mutual factorability at possibly different positive scales is equivalent to $S E_0 T$. The unscaled laws themselves are equal only when $S = T$. Every μ_S is an unscaled convolution factor of $\mu^{[2]}$.*

Proof. For each k cancellation of the common factors gives

$$\frac{A_{S,k}}{\gcd(A_{S,k}, A_{T,k})} = \prod_{\substack{j \leq k \\ j \in S \setminus T}} p_j.$$

The least common multiple of these initial products is finite exactly when $S \setminus T$ is finite, and is then the product in (29). Corollary 3.6 proves the equivalence and all the assertions about scales. Applying it in both directions gives finite symmetric difference. Finally equality of laws recovers equality of ladders by Proposition 3.7; the ratio at index j recovers membership of j in S . Taking $S = \emptyset$ in the unscaled criterion gives $\mu_\emptyset = \mu^{[2]} = \mu_T * \nu$ for every T , proving the final assertion. $\square$

Remark 5.2 (Orientation of the order). The displayed equations fix the order convention: the *target* μ_S contains a scaled copy of the *source* μ_T when $S \subseteq^* T$. Larger sets introduce more prime denominators and hence smaller summands. This avoids the ambiguity of calling convolution containment an order without specifying its direction.

5.2 The encoding is continuous in all derivatives

Equip $2^{\mathbb{N}_{>0}}$ with its product topology, and equip $C_{[-1,1]}^\infty(\mathbb{R})$ with the seminorms $\|f^{(r)}\|_\infty$, $r \geq 0$, restricting to functions supported in $[-1, 1]$.

Proposition 5.3 (A compact family of smooth densities). *The map $S \mapsto f_S$ is continuous and injective, and hence a homeomorphism onto a compact subset $\mathcal{K}$ of $C_{[-1,1]}^\infty(\mathbb{R})$. More explicitly, put*

$$\overline{A}_j = 2^j \prod_{i=1}^j p_i, \quad C_r = 2^{r+1} \prod_{j=0}^{r+1} \overline{A}_j.$$

If S, T agree on the first N positions and $N \geq r + 2$, then

$$\|f_S^{(r)} - f_T^{(r)}\|_\infty \leq C_r 2^{-N}. \quad (30)$$

Proof. Their prefix densities through index N agree. Couple the remaining tails using the same uniforms. Each tail is bounded by $1/(2A_{S,N})$, and $A_{S,N} = A_{T,N} \geq 2^N$. Therefore the difference of the two tails has absolute value at most $1/A_{S,N} \leq 2^{-N}$. The common prefix has r th derivative Lipschitz with constant $L_r(A_S) \leq C_r$, by the proof of Proposition 2.6. Taking expectations of the shifted prefix proves (30). This is continuity in every seminorm. Injectivity was proved in Theorem 5.1; a continuous injection from compact Cantor space into a Hausdorff space is a homeomorphism onto its image. $\square$

The common compact support and the continuity in every derivative are important. The classification barrier below is not produced by a sequence of increasingly singular target densities, by unbounded supports, or by a discontinuous coding map. It is present inside one compact family of ordinary smooth probability densities.

5.3 No Borel real-valued complete invariant

Define $f \sim g$ on $\mathcal{K}$ when each of the corresponding laws is a convolution of a positive dilation of the other with a probability measure. A real-valued *complete invariant* would be a map I such that $I(f) = I(g)$ holds exactly when $f \sim g$. The requirement that I be Borel measurable excludes arbitrary nonmeasurable choices of representatives.

Theorem 5.4 (Nonsmooth classification on a compact smooth family). *There is no Borel measurable real-valued complete invariant for $\sim$ on $\mathcal{K}$. There is also no complete invariant consisting of countably many Borel real-valued coordinates. In fact, every continuous real-valued invariant for $\sim$ is constant on $\mathcal{K}$.*

Proof. Suppose first that I were a Borel complete invariant, and pull it back along the continuous map of Proposition 5.3: $J(S) = I(f_S)$. By Theorem 5.1, J is unchanged under every finite modification of S .

Give the bits of S the independent fair-coin law. For each rational t , the Borel event $\{J \leq t\}$ is unchanged by modifying the first N bits, for every N . It is therefore measurable with respect to the bits after N : fix the first N bits to zero to see that its membership depends only on the remaining coordinates. Hence it is a tail event.

A tail event has probability zero or one. One elementary proof is that such an event is independent of each finite-coordinate cylinder; the monotone-class argument extends independence to the sigma-algebra generated by all cylinders, which contains the event itself. Its probability p then obeys $p = p^2$. It follows that the distribution function of the real random variable J takes only the values zero and one at rational arguments. Consequently J is almost surely equal to a constant: take the cut between rational arguments of probability zero and those of probability one. The cut is finite because J is real valued, and continuity of probability from above and below gives the claim.

Completeness of I would place this conull level set inside a single E_0 -class. But each such class is countable, since there are only countably many finite changes of a fixed sequence; each individual sequence has fair-coin measure zero. This is a contradiction.

For countably many Borel real coordinates, apply the same argument to each coordinate and intersect their conull sets of constancy. The complete coordinate vector is again constant on a conull set, giving the same contradiction. Finally each E_0 -class is dense in Cantor space,

since a finite change can prescribe any finite initial block. A continuous invariant constant on one such dense class is constant everywhere. $\square$

Remark 5.5 (What the obstruction does not say). The full Fourier transform remains a complete invariant for *equality* of probability laws. The theorem instead concerns the coarser relation of mutual scaled factorability. It does not say that this relation is non-Borel: on the constructed family it is the familiar Borel relation E_0 . Nor does it prohibit a nonmeasurable choice-based assignment of real labels. The precise negative conclusion is the absence of measurable complete labels by equality of real numbers or countable real sequences.

6 Effective classification: exact arithmetical complexity

6.1 Input model and the meaning of completeness

An effective assertion requires a presentation model. Use finite program codes for primitive-recursive functions specifying the positive ladder ratios, with validity guaranteed by construction. For example, from any primitive-recursive $h : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ take

$$A_0 = 1, \quad A_{k+1} = A_k(2 + h(k)).$$

This represents valid ladders without a separate undecidable totality or validity test. All the hardness constructions below have ratios of this form. The conclusions also hold as promise-problem statements for any larger presentation system containing these examples and providing their integer entries effectively.

A predicate is Π_1^0 when it has the form $\forall k R(e, k)$ with a decidable relation R ; it is Σ_2^0 when it has the form $\exists m \forall k R(e, m, k)$. Completeness means that every predicate at the indicated level can be transformed effectively into the problem under consideration, preserving yes and no answers. For the hardness proofs we use the usual primitive-recursive matrix normal forms, so the constructed ladder entries remain primitive recursive.

Theorem 6.1 (Effective complexity of exact factors). *For primitive-recursively presented divisibility ladders A, B :*

- (i) *For each fixed integer $m \geq 1$, existence of a factor in $\mu_A = D_{1/m} \mu_B * \nu$ is Π_1^0 -complete.*
- (ii) *Existence of a factor at some positive real scale c is Σ_2^0 -complete.*

The target and source densities in the hardness examples are uniformly computable together with all derivatives, with the quantitative error bounds of Proposition 2.6.

Proof. For (i), the upper bound follows from the criterion $\forall k (A_k \mid mB_k)$, which is a universal quantification of a decidable integer test.

For hardness, start with an arbitrary Π_1^0 predicate $\forall t R(e, t)$ in primitive-recursive normal form. Choose an odd prime p that does not divide the fixed integer m . Set $\varepsilon_0 = 0$ and, for $k \geq 1$, set $\varepsilon_k = 1$ if some $t < k$ violates $R(e, t)$, and $\varepsilon_k = 0$ otherwise. This is a nondecreasing binary sequence computed by a bounded search. Define

$$B_k = 2^k, \quad A_k = 2^k p^{\varepsilon_k}.$$

Both are valid primitive-recursive ladders. If there is no violation, they are equal and the factor exists. If there is a violation, all sufficiently late A_k contain the extra prime p , which cannot divide mB_k . The factor therefore exists exactly when the original Π_1^0 predicate holds.

For (ii), Theorem 3.2 eliminates quantification over arbitrary real scales and gives the Σ_2^0 upper bound

$$\exists m \geq 1 \forall k \geq 0 (A_k \mid mB_k).$$

To prove hardness, let W_e be a computably enumerable set of positive integers, with a fixed bounded-stage enumeration. Define

$$B_k = 2^k, \quad A_k = 2^k \prod_{\substack{1 \leq j \leq k \\ j \text{ enumerated in } W_e \text{ by stage } k}} p_j. \quad (31)$$

Membership in this finite stage condition is decided by bounded simulation; products and the enumeration of the first k odd primes are primitive recursive. The set of prime factors included is nondecreasing in k , so these are valid ladders. Every $j \in W_e$ eventually contributes p_j . Corollary 3.6 shows that some scale exists if and only if W_e is finite.

Here is a direct verification of the needed hardness of finiteness, rather than an unsupported appeal to it. Given $\exists n \forall t R(e, n, t)$ with primitive-recursive R , let ℓ_s be the least $n \leq s$ for which $R(e, n, t)$ holds for every $t \leq s$, or $s + 1$ if no such n exists. Enumerate a fresh integer whenever ℓ_s changes. If there is a true witness, the least one eventually survives while all smaller candidates have been refuted, so ℓ_s stabilizes and the enumerated set is finite. If there is no witness, every bounded collection of candidates is eventually refuted, so $\ell_s \rightarrow \infty$ and changes infinitely often. Thus the enumerated set is finite exactly when the given Σ_2^0 predicate holds. Composing this construction with (31) proves hardness.

Finally, every entry of the constructed ladders is uniformly computable from the input code. The rational spline and explicit derivative error bound in Proposition 2.6 apply uniformly to them. The complexity is therefore a property of exact factorization, not a failure to compute the smooth input functions. $\square$

6.2 Consequences for certificate-based computation

For a fixed reciprocal-integer scale, a search for a failed divisibility condition semidecides nonexistence: the first failed index is an exact certificate. A general terminating algorithm for both existence and nonexistence would contradict Theorem 6.1. Likewise, searching for successively better numerical agreement between transforms cannot solve the unrestricted exact problem.

This does not obstruct useful algorithms on structured subclasses. For constant integer bases, Theorem 4.2 reduces everything to finite prime factorization. More generally, a symbolic description that proves eventual periodicity or bounded prime deficits can certify the infinite condition. The theorem identifies what additional structural information an implementation must exploit; it does not say that every individual instance is difficult.

There is also a separation between computing a factor and recognizing that it exists. Once the arithmetic hypotheses are provided with a proof, (13) computes the factor weakly by its finite digit prefixes with an explicit tail bound. Such a computation is perfectly compatible with undecidability of the unpromised existence problem.

6.3 A terminating algorithm for eventually periodic ratios

Proposition 6.2 (Periodic-ratio decision theorem). *Suppose the integer ratios A_{k+1}/A_k and B_{k+1}/B_k are periodic for $k \geq K$ with a common period $P \geq 1$. Put*

$$Q_A = A_{K+P}/A_K, \quad Q_B = B_{K+P}/B_K.$$

Some positive-scale factor exists if and only if $Q_A \mid Q_B$. When this holds,

$$L(A, B) = \text{lcm}_{0 \leq k < K+P} \frac{A_k}{\gcd(A_k, B_k)}. \quad (32)$$

Thus the existence problem, all scales, and the largest scale are decided by a finite calculation from a certified eventually periodic presentation.

Proof. For $0 \leq r < P$ and $j \geq 0$, periodicity gives $A_{K+jP+r} = A_{K+r}Q_A^j$ and $B_{K+jP+r} = B_{K+r}Q_B^j$. At any prime p the exponent difference is therefore

$$v_p(A_{K+r}) - v_p(B_{K+r}) + j(v_p(Q_A) - v_p(Q_B)).$$

A positive slope makes the deficits unbounded and excludes every finite m . When all slopes are nonpositive, their maxima are attained at $j = 0$; together with the initial indices these are exactly the finite tests in (32). Corollary 3.6 completes the proof. $\square$

For example, alternating ratios (2, 6) and (3, 4) have the same period product 12. Their laws are mutually scaled-factorable, with minimal scale denominators 2 and 3 in the respective directions. On the other hand, period products 12 and 18 do not admit the factor in the first direction: ordinary numerical size $12 < 18$ is insufficient because $12 \nmid 18$. This finite algorithm and the unrestricted impossibility result apply to different input classes and are entirely consistent.

7 Same-base remainders: singularity and finite smoothing

7.1 Digit representation and extraction of full base factors

Fix integers $a \geq 2$ and $m \geq 1$. Let $\eta_{a,m}$ be the unique measure such that

$$\mu^{[a]} = D_{1/m}\mu^{[a]} * \eta_{a,m}. \quad (33)$$

Theorem 3.2 gives

$$\eta_{a,m} = \text{Law} \left(\frac{1}{m} \sum_{k \geq 0} a^{-k} \left(J_k - \frac{m-1}{2} \right) \right), \quad J_k \text{ uniform on } \{0, \dots, m-1\}. \quad (34)$$

The support radius is $(1 - 1/m)a/(2(a-1))$. For $m = 1$ this is δ_0 .

Write

$$m = a^d r, \quad d = \max\{j \geq 0 : a^j \mid m\}, \quad a \nmid r. \quad (35)$$

When $r = 1$, the last condition holds automatically because $a \geq 2$. For composite a , d is divisibility by powers of the *whole* base; it is not the valuation at a single arbitrarily chosen prime.

Proposition 7.1 (Uniform smoothing decomposition). *With (35),*

$$\eta_{a,m} = \left(\bigstar_{j=0}^{d-1} \mathbf{U}_{a^{-j}} \right) * D_{a^{-d}} \eta_{a,r}, \quad (36)$$

where an empty convolution is δ_0 .

Proof. The self-factor formula for the scale a^{-d} is obtained by splitting the geometric series into its first d terms and its tail. Thus $\eta_{a,a^d} = \bigstar_{j < d} \mathbf{U}_{a^{-j}}$. Apply the composition identity (18) with scales a^{-d} and $1/r$. This gives (36) without division at Fourier zeros. Equivalently, the quotient of transforms telescopes through the first d sinc factors. $\square$

7.2 The primitive remainder is singular continuous

Define the centered digit mask

$$h_r(u) = \frac{1}{r} \sum_{j=0}^{r-1} e^{2\pi i u(j-(r-1)/2)}. \quad (37)$$

The absolute value $|h_r|$ is one-periodic. The mask itself equals $h_r(u) = \sin(\pi r u)/(r \sin(\pi u))$ whenever the denominator is nonzero, with the removable values filled in by (37). Thus

$$\hat{\eta}_{a,r}(t) = \prod_{k \geq 0} h_r\left(\frac{t}{ra^k}\right). \quad (38)$$

Lemma 7.2 (Exact Fourier resonance). *If $r \geq 2$ and $a \nmid r$, the number*

$$C_{a,r} := \prod_{\ell \geq 1} \left| \frac{\sin(\pi r/a^\ell)}{r \sin(\pi/a^\ell)} \right| \quad (39)$$

is positive, and for every $N \geq 0$,

$$|\hat{\eta}_{a,r}(ra^N)| = C_{a,r}. \quad (40)$$

Proof. At $t = ra^N$, the mask arguments for $k \leq N$ are integers and their absolute values are one. Reindexing the remaining factors gives (39). Its denominators are nonzero. A numerator could vanish only if $a^\ell \mid r$, which would imply $a \mid r$. Every factor is therefore nonzero. As $\ell \rightarrow \infty$ the factor is $1 + O(a^{-2\ell})$, so the product of their absolute values converges to a nonzero number. $\square$

Theorem 7.3 (Primitive digit remainder). *If $r \geq 2$ and $a \nmid r$, the probability measure $\eta_{a,r}$ is atomless and singular with respect to Lebesgue measure. If $r < a$, its self-similar support has disjoint first-level pieces. If $r \geq a$, its support is its entire closed support interval, despite the singularity.*

Proof. The random series satisfies the self-similar distributional equation

$$Y \stackrel{d}{=} \frac{Y'}{a} + \frac{J - (r-1)/2}{r}, \quad (41)$$

with Y' an independent copy and J a uniform digit. This equation has a unique probability solution. Indeed, iterate its characteristic-function equation n times; the remaining characteristic factor is evaluated at t/a^n and tends to one. The other factors converge to (38), determining the law uniquely. This is a direct special-case proof of the contracting self-similar measure construction associated with [3].

The affine maps in (41) preserve absolute continuity and also preserve singularity of measures with respect to Lebesgue measure. The operator that averages their pushforwards therefore preserves both parts of the Lebesgue decomposition separately. If the stationary measure had both parts with positive mass, each part after normalization would be a stationary probability, contradicting uniqueness. The measure is consequently of pure type. Its Fourier transform does not tend to zero by Lemma 7.2, whereas the Fourier transform of every integrable density tends to zero. The latter elementary Riemann–Lebesgue assertion follows by approximating the density in L^1 by step functions on bounded intervals, whose Fourier integrals tend to zero explicitly. The measure cannot be absolutely continuous, and hence is purely singular.

To exclude atoms, suppose the largest atom mass were $p > 0$. Such a maximum is attained: only finitely many atoms can have mass at least half the supremum. Let E be the finite

nonempty set of atoms of mass p . For $x \in E$, the average in (41) can equal p only if every inverse image under all r affine maps also has mass p . Each inverse map is injective and sends E into E , hence permutes it. An affine map with expansion factor $a > 1$ cannot permute a finite set of positive diameter, so E is a singleton. Distinct digit maps have distinct fixed points and cannot all preserve that singleton. This contradiction proves atomlessness.

Finally the convex hull of the support has length $H = (r - 1)a/(r(a - 1))$. Its first-level images have length H/a and adjacent left endpoints separated by $1/r$. They are disjoint when $r < a$ and cover the full hull when $r \geq a$. In the latter case the hull is an invariant compact set, so it equals the self-similar attractor. Uniqueness of that compact attractor follows by iterating the maps: their Hausdorff contraction ratio is $1/a$. Alternatively, the possible digit prefixes plus a tail of diameter Ha^{-n} converge to the same set. $\square$

7.3 Sharp differentiability of every same-base remainder

The space $W^{d,1}(\mathbb{R})$ consists of integrable functions whose distributional derivatives through order d are integrable functions. A function is of bounded variation when its distributional first derivative is a finite signed measure. These distinctions matter for a continuous function whose derivative is singular, as with a Cantor distribution function.

Theorem 7.4 (Complete integer regularity classification). *Write $m = a^d r$ as in (35).*

- (i) *If $m = 1$, then $\eta_{a,m} = \delta_0$.*
- (ii) *If $d = 0$ and $r \geq 2$, then $\eta_{a,m}$ is singular continuous.*
- (iii) *If $d \geq 1$ and $r = 1$, then $\eta_{a,m}$ is the convolution of the d uniforms in (36). For $d = 1$ its density has no continuous representative. For $d \geq 2$ its density is C^{d-2} but not C^{d-1} . In every case it has bounded weak derivatives through order $d - 1$.*
- (iv) *If $d \geq 1$ and $r \geq 2$, then $\eta_{a,m}$ has a compactly supported density $f \in C^{d-1}(\mathbb{R})$. The derivative $f^{(d-1)}$ is continuous and of bounded variation, but is not absolutely continuous. Consequently $f \notin W^{d,1}(\mathbb{R})$ and $f \notin C^d(\mathbb{R})$.*

Proof. The first two assertions follow immediately from the digit representation and Theorem 7.3. For (iii), repeated convolution of interval indicators gives a degree- $(d - 1)$ spline, with continuous derivatives through order $d - 2$ when $d \geq 2$. Differentiate $d - 1$ different uniform factors distributionally and leave the last uniform undifferentiated; this also shows that the $(d - 1)$ st weak derivative is bounded. Near the left endpoint L of the support, before any other spline knot, the density is a positive constant times $(x - L)_+^{d-1}$. Its $(d - 1)$ st derivative has a nonzero jump, proving the sharp failure of C^{d-1} . For $d = 1$ this is exactly the jump of a uniform density.

For (iv), put $\lambda = D_{a^{-d}} \eta_{a,r}$, which is atomless singular. After differentiating $d - 1$ of the uniform factors, $f^{(d-1)}$ is a finite linear combination of translates of the convolution of one uniform with λ . Such a convolution is continuous: for a uniform of length ℓ its density is

$$x \mapsto \frac{\lambda([x - \ell/2, x + \ell/2])}{\ell},$$

with endpoint conventions immaterial because λ is atomless. Its continuity follows from continuity of the distribution function of λ . This proves $f \in C^{d-1}$.

Differentiating all d uniform factors instead gives

$$D^d f = \left(\bigast_{j=0}^{d-1} a^j (\delta_{-a^{-j}/2} - \delta_{a^{-j}/2}) \right) * \lambda, \quad (42)$$

a finite signed sum of translates of a singular measure. It is a singular finite signed measure. It is nonzero, since $D^d f = 0$ would make f a polynomial of degree at most $d - 1$ as a distribution, and a compactly supported polynomial must be zero, contradicting mass one. Thus $f^{(d-1)}$ is of bounded variation but cannot be absolutely continuous: its derivative is a nonzero singular measure. The asserted Sobolev and classical nonmembership follow. $\square$

Base a	m	Remainder type and sharp integer regularity
2	2	One uniform; discontinuous density.
2	3	Singular continuous with interval support.
2	4	Two-uniform spline; continuous, not C^1 .
2	6	Continuous density with a nonzero singular derivative.
2	8	Three-uniform spline; C^1 , not C^2 .
2	12	C^1 density whose second derivative is singular.
3	2	Separated two-digit Cantor-type measure.
3	6	Continuous density of exact Hölder exponent $\log 2 / \log 3$.

The last entry uses the sharper result proved in Section 8.2. The table concerns the remainder, not the target: all geometric target laws in it have C^∞ densities.

8 Sharp Fourier order and separated-digit Hölder regularity

8.1 The exact power envelope after smoothing

The density regularity in Theorem 7.4 has a matching Fourier statement. The nondecaying primitive digit transform loses exactly one power for each extracted uniform factor, not an unspecified amount.

Lemma 8.1 (Offset resonance). *Let $r \geq 2$, $a \nmid r$, and $\delta \in \mathbb{R}$. Then*

$$\lim_{N \rightarrow \infty} |\widehat{\eta}_{a,r}(ra^N + \delta)| = C_{a,r} |\widehat{\eta}_{a,r}(\delta)|. \quad (43)$$

Proof. Split the mask product (38) at $k = N$. For $k \leq N$, periodicity of the absolute value of h_r gives

$$\left| h_r \left(a^{N-k} + \frac{\delta}{ra^k} \right) \right| = \left| h_r \left(\frac{\delta}{ra^k} \right) \right|.$$

Their product converges to $|\widehat{\eta}_{a,r}(\delta)|$. For $k = N + \ell$ with $\ell \geq 1$, the factors are

$$\left| h_r \left(a^{-\ell} + \frac{\delta}{ra^{N+\ell}} \right) \right|.$$

For each fixed ℓ they tend to $|h_r(a^{-\ell})|$. The convergence of the infinite products is justified uniformly in N after discarding finitely many initial values of N : for large ℓ , all arguments are bounded by a constant times $a^{-\ell}$ and $h_r(u) = 1 + O(u^2)$ near zero. The summable bound $O(a^{-2\ell})$ controls the product tails uniformly. The limiting tail product is $C_{a,r}$. $\square$

Theorem 8.2 (Sharp power Fourier decay). *Let $m = a^d r$ as in (35), with $d \geq 1$. Then*

$$|\widehat{\eta}_{a,m}(t)| = O(|t|^{-d}) \quad (|t| \rightarrow \infty), \quad \limsup_{t \rightarrow \infty} t^d |\widehat{\eta}_{a,m}(t)| > 0. \quad (44)$$

More precisely, let $t_N = ma^N + 1/2$. With the conventions $C_{a,1} = 1$ and $\hat{\eta}_{a,1} = 1$,

$$\lim_{N \rightarrow \infty} t_N^d |\hat{\eta}_{a,m}(t_N)| = \frac{a^{d(d-1)/2}}{\pi^d} \left(\prod_{j=0}^{d-1} \sin \frac{\pi}{2a^j} \right) C_{a,r} \left| \hat{\eta}_{a,r} \left(\frac{1}{2a^d} \right) \right| > 0. \quad (45)$$

Thus the power d cannot be replaced by any larger power in a global upper bound. When $d = 0$ and $r \geq 2$, there is no decay to zero at all.

Proof. Taking Fourier transforms in (36) gives

$$\hat{\eta}_{a,m}(t) = \left(\prod_{j=0}^{d-1} \text{sinc}(t/a^j) \right) \hat{\eta}_{a,r}(t/a^d).$$

The last factor has modulus at most one, and the finite product is $O(|t|^{-d})$. For the special sequence t_N , every ma^N/a^j with $j < d$ is an integer. Consequently

$$t_N^d \prod_{j=0}^{d-1} |\text{sinc}(t_N/a^j)| = \frac{a^{d(d-1)/2}}{\pi^d} \prod_{j=0}^{d-1} \sin \frac{\pi}{2a^j}.$$

The remaining argument is $t_N/a^d = ra^N + 1/(2a^d)$. Apply Lemma 8.1, or use the constant transform if $r = 1$, to obtain (45). Every finite sine factor is positive. The residual transform at $\delta = 1/(2a^d)$ is nonzero: in (38) its numerator arguments are $\delta/a^k \in (0, 1)$, no factor vanishes, and the tail product is nonzero. Finally the $d = 0$ assertion is Lemma 7.2. $\square$

A positive limsup is the appropriate sharpness statement, not a pointwise lower bound: these transforms have many zeros. In particular, the theorem does not assert $|\hat{\eta}(t)| \asymp |t|^{-d}$ at every large frequency.

8.2 An exact fractional exponent in the separated case

When $2 \leq r < a$, let $\alpha = \log r / \log a \in (0, 1)$. For a function g , Hölder continuity of exponent α means $|g(x) - g(y)| \leq C|x - y|^\alpha$ for all real x, y and a fixed constant C .

Theorem 8.3 (Sharp separated-digit Hölder exponent). *Assume $2 \leq r < a$. The distribution function of $\eta_{a,r}$ is globally Hölder continuous of exponent $\alpha = \log r / \log a$, and is not Hölder continuous with any exponent β satisfying $\alpha < \beta \leq 1$. For $d \geq 1$ and $m = a^d r$, the density in Theorem 7.4 satisfies*

$$f \in C^{d-1,\alpha}(\mathbb{R}), \quad f \notin C^{d-1,\beta}(\mathbb{R}) \quad (\alpha < \beta \leq 1). \quad (46)$$

Here $C^{d-1,\alpha}$ means that the $(d-1)$ st derivative is globally Hölder of exponent α .

Proof. Let H be the convex-hull length of the primitive support. At level n , there are r^n cylinder sets, each of mass r^{-n} , each contained in an interval of length Ha^{-n} . Strong separation gives a constant $g > 0$ such that distinct level- n hull intervals are separated by at least $ga^{-(n-1)}$. Indeed, two cylinders first differing at level $j \leq n$ lie in different first-level pieces of a common level- $(j-1)$ cylinder; scaling the first-level gap gives at least this bound.

An interval of length at most Ha^{-n} can therefore meet only a bounded number of level- n cylinders, independently of n . Choose n so that $Ha^{-(n+1)} < |I| \leq Ha^{-n}$. Its measure is at most Kr^{-n} for a constant K , and $r^{-n} = (a^{-n})^\alpha \leq (a|I|/H)^\alpha$. The finitely larger interval sizes are absorbed by increasing the constant. Thus $\eta_{a,r}(I) \leq C|I|^\alpha$, which proves the Hölder bound for its distribution function.

For sharpness let L be the left endpoint of the primitive support. The leftmost level- n cylinder lies in $[L, L + Ha^{-n}]$ and has exactly mass r^{-n} ; no other cylinder intersects this hull. Since there is no atom at L , the distribution-function increment on this interval is r^{-n} . The quotient of that increment by $(Ha^{-n})^\beta$ is $H^{-\beta}a^{n\beta}/r^n$, which tends to infinity for $\beta > \alpha$.

For the smoothed density set $\lambda = D_{a^{-d}\eta_{a,r}}$, $\ell_j = a^{-j}$, and $R_d = \frac{1}{2} \sum_{j < d} \ell_j$. Integrating (42) once from $-\infty$ gives the explicit formula

$$f^{(d-1)}(x) = \frac{1}{\prod_{j < d} \ell_j} \sum_{E \subseteq \{0, \dots, d-1\}} (-1)^{|E|} F_\lambda \left(x + R_d - \sum_{j \in E} \ell_j \right). \quad (47)$$

Each summand is Hölder of exponent α , proving the upper assertion. Near the left endpoint of the support of f , with distance smaller than $\min_j \ell_j$, all terms except $E = \emptyset$ vanish. At that endpoint the sharp increments just established for the scaled distribution function therefore survive without cancellation, proving the lower assertion. $\square$

The separated assumption is substantive. When $r \geq a$ and $a \nmid r$, the measure is still singular by Theorem 7.3, but cylinder intervals overlap and the bounded-intersection estimate used here is no longer valid. No formula for its exact Hölder exponent is asserted.

9 Verification, mathematical dependencies, and integration

9.1 What was checked computationally

The accompanying `verify.py` uses only Python's standard library, unbounded integers, and `Fraction`. Its exact finite tests were run successfully for this article. The machine-readable receipt is `verification_results.json`. The principal test counts are:

Test family	Cases or equalities
Five-entry ladders with ratios in $\{2, 3, 4\}$, all ordered pairs, scale denominators 1 through 12	78,732
Additional zero-order comparisons at fixed integer frequencies	214,180
Exact uniform interval tilings	96
Uniform splitting moments, through order 12	1,248
Geometric finite-prefix factorization cases	108
Finite-prefix factorization moments, through order 10	1,188
Ordered pairs of subsets of an eight-element prime index set	65,536
Allowed integer-base/scale triples checked at initial powers	920
Forbidden integer-base/scale triples with explicit prime witnesses	6,300
Composition-of-remainders moment equalities, through order 10	825
Period-two ratio schedules and scale denominators 1 through 24	6,144

The ladder tests compare coordinate divisibility, zero-order conditions at source generators, and the lcm criterion. The moment tests check the positive digit construction independently through exact rational moments. For forbidden integer-base pairs the program computes a prime and an index at which the zero-order inequality fails; it does not infer impossibility from a finite absence of successful examples.

These are regression tests for algebra, indexing, normalization, and finite instances. They are *not* a replacement for the proofs of infinite product convergence, singularity, Borel nonsmoothness, or hierarchy completeness. No random sampling or floating-point experiment is used as evidence for those conclusions.

9.2 Proof dependencies and candidate formalization boundaries

The mathematical dependencies can be separated into four layers. The first is elementary integer arithmetic: divisibility chains, prime valuations, initial-segment zero counts, and lcm obstructions. The second is probability: independent bounded sums, finite uniform splitting, and uniqueness of characteristic functions. The third is analysis: entire product convergence, zero multiplicities, Fourier inversion, and finite-measure distributional derivatives. The fourth is classification: prime coding, tail zero-one arguments, and explicit computability reductions.

The repository’s existing product and probability infrastructure is a natural place to reuse established interfaces [5]. A sound formalization plan should not simply postulate that a quotient is a characteristic function. It should first formalize the uniform splitting identity and assemble the actual independent digit law. Conversely, the necessary direction should record compactness of the unknown factor before using entire zero multiplicities.

A practical first formalization target is the arithmetic equivalence between $\forall k (A_k \mid mB_k)$ and the generator zero-count inequalities. A second target is the finite-prefix probability identity, followed by its limit under an explicit deterministic tail bound. Those components already isolate most of the main theorem’s logical burden. The periodic-ratio algorithm then reduces to finite arithmetic. The non-Borel classification and hierarchy-completeness sections require additional libraries and should not be labeled verified merely because the arithmetic component compiles.

9.3 Result-status ledger

Result or ingredient	Status in this article
Dyadic sinc product and dyadic zero orders	Classical; attributed to Arias de Reyna.
Dyadic identical-convolution rootlessness	Identified repository overlap; included only as a baseline corollary.
General ladder and cross-base factor classifications	Proved here from explicit hypotheses; broader priority unconfirmed.
Prime coding in a compact smooth family; effective hardness	Proved here; classical E_0 and computability methods are not claimed as new.
Primitive singularity and smoothing classification	Full proofs supplied; related to classical self-similar-measure theory, with no exhaustive novelty claim.
Sharp Fourier order and separated Hölder exponent	Proved here, including noncancellation and resonance details.
Finite verification program	Executed successfully; not a Lean proof.
Further research questions	Proposed boundaries and extensions, not assertions that their global open status has been fully audited.

10 Further research questions and topics

The questions below specify extensions not settled by this article. They should be checked against the relevant specialist literature before being advertised as globally open problems. The aim is to identify concrete next theorems and the precise mechanisms that the present proofs leave out.

Research question 1 (Beyond divisibility ladders). For summable positive lengths (ℓ_k) and (s_k) , characterize when $\prod_k \text{sinc}(\ell_k t) / \prod_k \text{sinc}(c s_k t)$ is the Fourier transform of a probability measure. Which additional hypotheses make zero-divisor domination sufficient, and can one construct a compactly supported uniform-product counterexample where all zeros cancel but positivity fails?

Theorem 3.2 uses an initial-segment property and coordinatewise interval tilings. Without a ladder, a successful approach may require matching several source factors to one target block, or a genuinely new positive-factor criterion. A useful first target is a finite union of integer divisibility chains with specified interlacing.

Editorial note (ProveIt, 2026-09-29). For a geometric target $\mu^{[p^h]}$ with p prime and an arbitrary summable family of uniform sources, the later unreviewed draft `Simultaneous_Convolution_Divisors_Fabius_Type_Laws/` of this group (2026-09-29) proves that zero-divisor domination is sufficient: it is equivalent to a nested matching of source factors to target scales and to factorization by a probability measure. Its base-six theorem gives the requested counterexample: in the notation of this article, $\prod_{k \geq 0} \text{sinc}(t/6^k) / (\text{sinc}(t/2) \text{sinc}(t/3))$ extends to an entire function that is not the Fourier transform of a positive measure, so $U_{1/2} * U_{1/3}$ does not divide $\mu^{[6]}$ although all zeros cancel. That draft writes $X_p = \sum_k p^{-k} U_k$ with U_k uniform on $[-1, 1]$, so its X_p is twice a variable with law $\mu^{[p]}$ here; it does not cite this article, whose one-ladder theorems are single streams of its results. Targets that are not prime powers remain open.

Editorial note (ProveIt, 2026-09-30). The later unreviewed draft `Arithmetic_Rigidity_off_Resonance_Geometric_Uniform_Laws/` of this group (batch 69 of the repository's docs/incoming/, 2026-09-30), written in the normalization of this article, gives a second hypothesis under which zero-divisor domination is sufficient. If the target lengths ℓ_k are pairwise rationally incommensurable, a finite or countable summable family of uniforms divides the law of $\sum_k \ell_k U_k$ exactly when the Fourier quotient is entire, and exactly when the uniforms have lengths $\ell_{\kappa(j)}/n_j$ for an injective assignment κ and integers $n_j \geq 1$ (its `thm:separated`). Its `thm:channels` gives the same conclusion for the geometric targets μ_p with $p = r^{-e/h}$, r prime, through h independent prime-adic Hall conditions.

Research question 2 (Two arbitrary geometric ratios). Determine all triples (p, q, c) with $0 < p, q < 1$, $c > 0$, for which $\mu_p = D_c \mu_q * \nu$ has a probability solution.

Theorem 4.1 treats $p = q$, and Theorem 4.2 treats $p = 1/a$ for an integer a . For general p , the target zeros are no longer confined to integers. The necessary inclusion of multiplicative zero sets suggests a role for relations between powers of p and q , but zero inclusion alone has not been proved sufficient in that setting.

Editorial note (ProveIt, 2026-09-30). The later unreviewed draft `Arithmetic_Rigidity_off_Resonance_Geometric_Uniform_Laws/` of this group (batch 69, 2026-09-30; same normalization) answers this question for every target ratio p no positive power of which is rational, a co-countable set: $\mu_p = D_c \mu_q * \nu$ has a probability solution exactly when $c = p^a/n$ and $q = p^m/d$ with integers $a \geq 0$ and $m, n, d \geq 1$ (its `thm:geometric`). For $p = r^{-e/h}$ with r prime and $\gcd(e, h) = 1$, its `thm:orbit` gives a finite criterion: the least $\ell \geq 1$ with q^ℓ rational exists and divides h , the widths cq^j , $0 \leq j < \ell$, lie in distinct rationality classes $p^s \mathbb{Q}$, $0 \leq s < h$, and

$q^\ell = 1/D$ for an integer D divisible by r^e . In both cases zero inclusion is sufficient. Together with Theorem 4.2, the question remains open only for targets whose least rational power p^h is A/B in lowest terms with $A > 1$, or $1/B$ with $h \geq 2$ and B not a prime power. That draft re-proves Theorem 4.1 as its `prop:same`, without citing this article.

Research question 3 (Regularity of cross-base digit remainders). When $a \mid b$ and $b > a$, classify the absolute continuity, exact smoothness, and Fourier decay of $\nu_{(a^k), (b^k), m}$.

Here digit counts grow as $m(b/a)^k$, so the remainder is not the fixed-digit self-similar law treated in Sections 7 and 8. The finite digit mesh becomes much smaller than the width it fills. A quantitative comparison with an infinite uniform product might yield smoothing beyond the same-base classification, but this requires a proof controlling all frequency scales.

Research question 4 (Overlapping primitive digits). For $r \geq a$ with $a \nmid r$, find effective formulas or algorithms for local dimensions and optimal Hölder exponents of the distribution function of $\eta_{a,r}$.

The measure has interval support and is singular; neither fact determines its local exponents. Finite carry-state descriptions of base- a expansions offer one concrete route. The separated-case exponent $\log r / \log a$ cannot simply be reused: it is at least one in this range and the separated-cylinder proof breaks down.

Research question 5 (Finer function-space regularity). Determine sharp Sobolev, Besov, and fractional-variation membership for the densities of $\eta_{a,a^d r}$, especially when the primitive digits overlap.

The article fixes the last continuous integer derivative and proves a sharp pointwise power Fourier envelope. These do not alone determine integrated Fourier norms or all fractional smoothness scales, because a resonant subsequence can be thin. Quantifying the width and frequency of resonance neighborhoods is a possible next step.

Research question 6 (Optimal approximate factorization). Determine or sharply estimate $\inf_\nu W_1(\mu_A, D_{1/m} \mu_B * \nu)$, and identify when the Fourier obstruction in Proposition 3.9 is asymptotically optimal.

The proposition already gives a strictly positive, support-independent lower bound from any failed divisibility condition. Remaining targets include upper bounds from constructive approximate factors, optimization over witness frequencies, and analogous sharp estimates in other metrics. Any estimate must depend on the witness scale: Proposition 5.3 shows how small changes in high prime positions can be in every fixed derivative norm. A uniform separation over all nonfactorable pairs is therefore not a plausible substitute for a scale-sensitive analysis.

Editorial note (ProveIt, 2026-09-30). For geometric targets off resonance, the batch-69 draft named in the notes after “Beyond divisibility ladders” and “Two arbitrary geometric ratios” bounds $\inf_\nu W_1(\mu_q, U_{1/N} * U_{1/M} * \nu)$ from below by the mechanism of Proposition 3.9 (its `thm:metric`, which does not cite this article). For $U_{1/2} * U_{1/3}$ it shows that the infimum is 0 at $q = 1/2$, positive at every q no positive power of which is rational, and at most $|q - 1/2|/(2(1 - q))$ (its `thm:perturbation`), so exact factorability is lost under arbitrarily small changes of q while the distance tends to zero. The sharp rate is open there too.

Research question 7 (Decidable presentations beyond eventual periodicity). Extend Proposition 6.2 to ratio sequences given by finite automata, substitutions, or other certified finite-state presentations.

The arithmetic problem is to decide whether all prime-exponent deficits remain bounded above and whether only finitely many primes contribute. For a finite alphabet of ratios, the second issue disappears, leaving a finite collection of weighted prefix-sum boundedness problems. The precise answer depends on the presentation class and should be stated with its input model and complexity, rather than as a generic algorithm for all computable ladders.

Research question 8 (Classification complexity on larger classes). What is the exact Borel classification complexity of mutual scaled convolution factorability for arbitrary smooth compactly supported probability laws, or for broader infinite uniform products?

Theorem 5.4 supplies a concrete E_0 lower bound, even on a compact family. It does not establish a maximal lower bound or an upper classification theorem for the ambient class. A useful intermediate problem is to determine the complexity for finitely many interlaced ladders.

Research question 9 (Multivariate arithmetic factors). Develop an analogue of Theorem 3.2 for sums of uniforms on nested boxes, parallelotopes, or lattice fundamental domains in $\mathbb{R}^d$.

For products of independent one-dimensional ladders, the theorem applies coordinatewise when the dilation is diagonal. Non-diagonal integer maps introduce genuine lattice geometry. Smith normal forms may describe finite uniform tilings, but the Fourier zero sets are hyperplanes rather than isolated points, and multiplicity bookkeeping must be redesigned.

Editorial note (ProveIt, 2026-09-29). For the independent product of prime-base geometric targets $\mu^{[p]^{\otimes d}}$, the later unreviewed draft `Simultaneous_Convolution_Divisors_Fabius_Type_Laws/` of this group (2026-09-29) proves that a sum of uniform laws along arbitrary vectors in $\mathbb{R}^d$ is a convolution factor exactly when every generator is coordinate-aligned with a signed reciprocal-integer length and each coordinate satisfies the one-dimensional matching criterion, and that the real matrices C for which $C_*\mu^{[p]^{\otimes d}}$ is a convolution factor of $\mu^{[p]^{\otimes d}}$ are exactly the partial monomial matrices with nonzero entries $\pm 1/n$. Nested boxes, parallelotopes and lattice fundamental domains in general remain open.

Research question 10 (Certified probability-factor software). Implement an exact factorization engine that takes a certified ladder presentation, returns either an explicit zero-order obstruction or a proof of the infinite divisibility condition, and produces a verified sampler or weak approximation for the remainder.

The unrestricted decision problem is impossible by Theorem 6.1, so the specification must permit an “undetermined” result outside certified subclasses. Positive outputs should include a constructive digit law, not only cancellation of an entire quotient. The finite-prefix test suite in this package is a regression starting point, while the convergence and positivity statements are the essential proof obligations.

11 Conclusion

The classification problem admits an unusually exact answer once its arithmetic structure is exposed. For nested integer denominators, probability factorization is equivalent to divisibility at every level; the remainder is then an explicit independent-digit sum. The same criterion simultaneously yields elementary cross-base rigidity, a finite algorithm for periodic schedules, and sharp nonclassification results for unrestricted presentations.

This coexistence is the main conceptual point. The targets are smooth, compactly supported, and effectively approximable, yet their exact convolution relation can remember infinitely many independent arithmetic choices. The remainder measures also retain that structure

analytically: removing a scaled smooth factor can leave a singular law, a finite spline, or a density whose highest continuous derivative has singular variation. The written proofs and finite exact checks establish these statements within the hypotheses given here; the broader questions and priority assessment remain separate research tasks.

A Reproduction and source notes

The package is self-contained for typesetting. From its directory, run

```
python3 verify.py --output verification_results.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The three TeX passes resolve the contents and cross-references. A `Makefile` is included as a convenience. The PDF supplied with this package was generated from this TeX source, not copied from the repository. No external figures, downloaded font files, proprietary packages, or network requests are required. Standard LaTeX packages listed in the preamble are needed.

`SOURCES.md` records the inspected repository locations and primary bibliographic sources. The repository snapshot is identified by the commit printed in Section 1; subsequent changes are outside this audit. The standalone article is an addition for consideration, not a commit, pull request, or modification of the user’s repository.

References

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